

Lesson 26: Standard Deviation & the GDC — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Does σ measure centre or spread?
2. σ when all values are equal?
3. Bigger σ means more or less consistent?
4. Which is more spread: $\sigma = 3$ or $\sigma = 11$?
5. Find the mean of 4, 6, 8.
6. Does adding 10 to every value change σ ?

Section 2 — Core practice

7. Set A: $\sigma = 1.5$; Set B: $\sigma = 9$. Which is more consistent?
8. Two classes have the same mean. A has $\sigma = 3$, B has $\sigma = 8$. Which is more consistent?
9. A data set has $\sigma = 0$. What does this tell you?
10. Does multiplying every value by 2 change the spread?
11. Mean 50, $\sigma = 4$ vs mean 50, $\sigma = 12$: which scored more consistently?

12. If every value increases by 7, what happens to the mean and σ ?

Section 3 — Challenge & exam-style

13. Class A: mean 64, $\sigma = 5$. Class B: mean 70, $\sigma = 5$. Compare them.

14. Class A: mean 60, $\sigma = 4$. Class B: mean 60, $\sigma = 13$. Compare them.

15. A test is marked out of 100. Group X: mean 55, $\sigma = 20$. Group Y: mean 55, $\sigma = 6$. Which is more reliable as a typical score?

16. Explain why standard deviation is preferred to range as a measure of spread.